COMMONWEALTH OF AUSTRALIA

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Michael Woolridge
Wiley, 2009.

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FIT5226
Multi-agent System & Collective Behaviour

Wk 12: Auctions (Resource Allocation)

Bernd Meyer, September 2024

Resource Allocation

- Allocation of scarce resources amongst a number of agents is central to multi-agent systems.

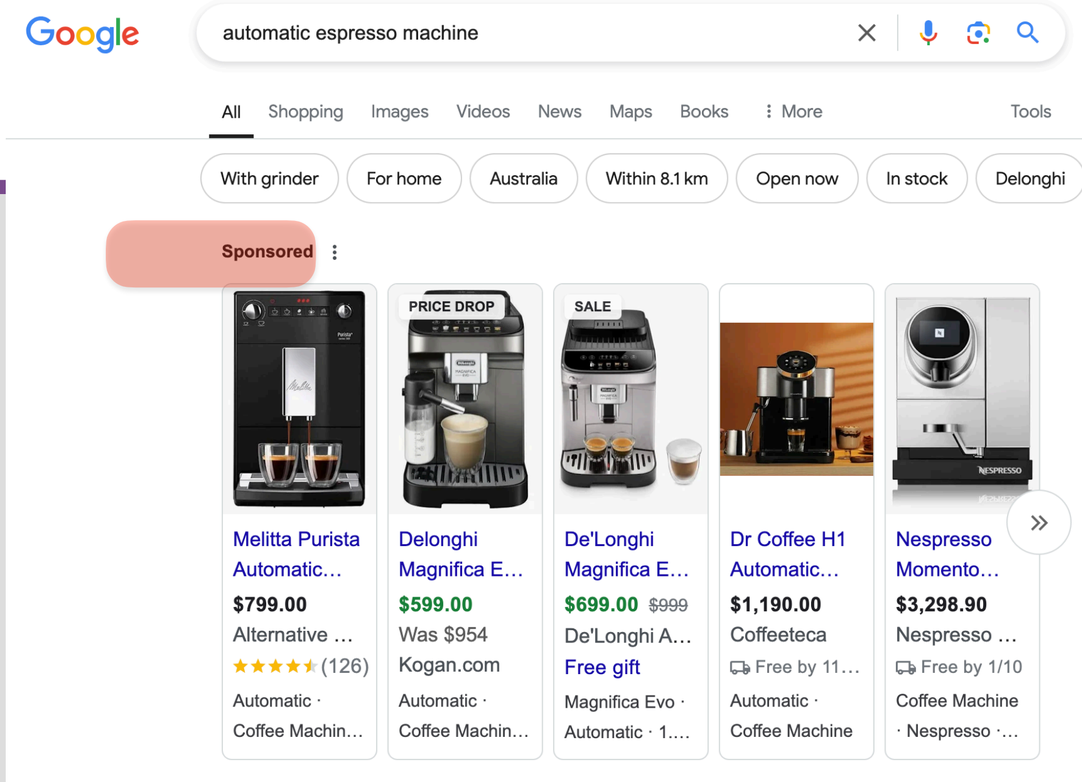
- A resource might be:
  - a physical object
  - the right to use land
  - computational resources (processor, memory, . . .)

- It is a question of supply vs demand
  - If the *resource isn't scarce*…, or if there is *no competition* for the resource...
    - ... there is no trouble allocating it
  - If there is a *greater demand than supply*
    - … we need to determine how to allocate it

Auctions

- In practice, this means we will be talking about auctions.
  - These used to be rare
  - However, auctions have grown massively with the Web/Internet
    - Frictionless commerce

- Now feasible to auction things that weren't previously profitable:
  - eBay
  - Adword auctions



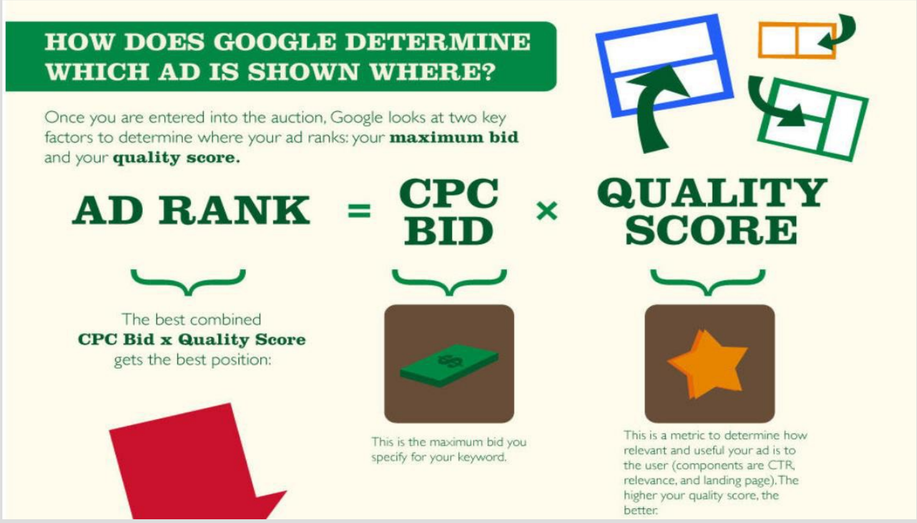


image from: https://www.wordstream.com/articles/what-is-google-adwords

*"... An **auction** is a market institution in which messages from traders include some price information — this information may be an offer to buy at a given price, in the case of a **bid**, or an offer to sell at a given price, in the case of an **ask** — and which gives priority to higher bids and lower asks..."*

This definition, as with all this terminology, comes from Dan Friedman (Professor at UCSC, US)

- An auction takes place between an agent known as the auctioneer and a collection of agents known as the bidders.
- The goal of the auction is for the auctioneer to allocate a 'good' (i.e. a scarce resource) to one of the bidders.
- The **auctioneer** desires to **maximize** the price at which the good is allocated, while **bidders** desire to **minimize** the selling price.

Types of Value

- What are types of value of the good?
    - Private Value
    - Common Value
    - Correlated Value

Private Value

Good has an value to me that is independent of what it is worth to you.

- *For example: John Lennon's last dollar bill.*

Common Value

The good has the same value to all of us.

Correlated Value

Our values are related (but not the same).

The more you're prepared to pay, the more I should be prepared to pay.

Auction Characteristics

- *Winner Determination*
  - Who gets the good, and what do they pay?
    - First price auction: the agent that bids the most is allocated the good, and pays the amount of the bid.
    - Second price auction: allocate the good to the agent that bid the highest, but this agent pays only the amount of the second highest bid.
- *Open Cry* vs *Sealed-bid*
  - Whether or not the bids made by the agents are known to each other?
    - Open cry: every agent can see what every other agent is bidding
    - Sealed-bid: the agents are not able to determine the bids made by other agents
- *One-shot* vs *Iterated Bids*
  - Is there a single round of bidding (i.e. one-shot) after which the good is allocated?
  - If multiple bids are permitted, then:
    - Does the price ascend, or descend for successive bids?
    - What is the terminating condition?

English Auction

- English auctions are the most commonly known type of auction, made famous by such auction houses as Sotheby's, Melbourne real-estate and more recently eBay.
- Buyers call out bids, bids increase in price.
  - In some instances the auctioneer may call out prices with buyers indicating they agree to such a price.
- The seller may set a *reserve price*, i.e., the lowest acceptable price.
- Auction ends:
  - at a fixed time (internet auctions); or when there is no more bidding activity.
  - The "last man standing" pays their bid.

What strategy should we use in English Auction?

- successively bid a small amount more than the current highest bid until the bid price reaches our private valuation, and then to withdraw.

English Auction

Classified in the terms we used above:

- ***First-price***
- ***Open-cry***
- ***Ascending***

Around 95% of internet auctions are of this kind. The classic use is the sale of antiques and artwork.

Susceptible to:

- *Winner's curse (the winner's bid exceeds the intrinsic value or true worth of an item.*
- *Shills*

Dutch Auction

- Also called a “descending clock” auction
  - Some auctions use a clock to display the prices.
- Starts at a *high price*, and the auctioneer calls out *descending prices*.
  - One bidder claims the good by indicating the current price is acceptable.
    - *Ties are broken* by restarting the descent from a slightly higher price than the tie occurred at.
- The winner pays the price at which they “stop the clock”.

Dutch Auction

Classified in the terms we used above:

- ***First-price***
- ***Open-cry***
- ***Descending***

High volume (since auction proceeds swiftly).
Often used to sell perishable goods:

- *Flowers in the Netherlands*
- *Fish in Spain and Israel.*
- *Tobacco in Canada.*

First-Price Sealed-bid Auction

- One-shot: there is a single round, in which each bidder submits to the auctioneer a bid for the good;
- The good is awarded to the agent that made the highest bid. The winner pays the price of the highest bid.
- In an English auction, you get information about how much a good is worth.
  - Other people's bids tell you things about the market.
- In a *sealed bid auction*, none of that happens
  - at most you know the winning price after the auction.

First-Price Sealed-Bid (FPSB)

Classified in the terms we used above:

- ***One-shot***
- ***First-price***
- ***Sealed Bid***

Governments often use this mechanism to sell treasury bonds (the UK still does, although the US recently changed to Second-Price sealed Bids).

Property can also be sold this way (as in Scotland).

Vickrey Auction

- The Vickrey auction is a one-shot and sealed bid auction.
  - The winning bid is the *highest bid*, but the winning bidder *pays the amount of the second highest bid*.
- This sounds odd, but it is actually a very smart design because it is *in the bidders' interest to bid their true valuation about the good*.
    - Vickrey auctions are not prone to strategic manipulation

Vickrey Auction

Classified in the terms we used above:

- ***One-shot***
- ***Sealed Bid***
- ***Second-price***

Historically used in the sale of stamps and other paper collectibles.

Why it works…

- Suppose you bid more than your valuation.
  - You may win the good.
  - If you do, you may end up paying more than the amount of your private valuation,
  - i.e. you may effectively may loose money
  - Not so smart!
- Suppose you bid less than your valuation.
  - You stand less chance of winning the good.
  - Moreover, even if you win, the amount you pay will not have been affected by the fact that you bid less than your true valuation, i.e., you will pay the second highest bid.
  - Not so smart!

Dominance of Truthful Bidding

- Let v_i be the bidder i's value for an item, and b_i be the bidder's bid
  - The payoff for bidder i is:

$$p_i = \begin{cases} v_i - max_{j \neq i} b_j & \text{if } b_i > max_{j \neq i} b_j \\ 0 & \text{otherwise} \end{cases}$$

- Assume bidder i bids $b_i > v_i$ (i.e. *overbids*)
  - If $max_{j \neq i}\, b_j < v_i$, then the bidder would win, regardless the bidder uses b_i or v_i for bidding, and the bidder will pay $max_{j \neq i}\, b_j$. Therefore the strategies of bidding truthfully and overbidding have equal payoffs
  - If $max_{j \neq i}\, b_j > b_i$, then the bidder would loose whether or not the bid was truthful. Again, both strategies have equal payoffs (e.g., 0)
  - If $v_i < max_{j \neq i}\, b_j < b_i$, then the strategy of overbidding would win the action and pay $max_{j \neq i}\, b_j$, but the payoff would be negative (because $v_i < max_{j \neq i}\, b_j$). A truthful strategy would pay zero (i.e., the bidder is not the winner).

Dominance of Truthful Bidding (cont'd)

- Let v_i be the bidding agent i's value for an item, and b_i be the agent's bid
  - The payoff for bidder i is:

$$p_i = \begin{cases} v_i - max_{j \neq i} b_j & \text{if } b_i > max_{j \neq i} b_j \\ 0 & \text{otherwise} \end{cases}$$

- Assume bidder i bids $b_i < v_i$ (i.e. *underbids*)
  - If $max_{j \neq i}\, b_j > v_i$, then the bidder would loose regardless of whether the bidder uses b_i or v_i for bidding. Therefore the strategies of bidding truthfully and underbidding have equal payoffs
  - If $max_{j \neq i}\, b_j < b_i$, then the bidder would win whether or not the bid was truthful. Again, both strategies have equal payoffs
  - If $b_i < max_{j \neq i}\, b_j < v_i$, then only the strategy of signalling truthfully would win the action, with a positive payoff. Underbidding strategy pays zero.

Collusion

- None of the auction types discussed so far are immune to collusion
  - A *grand coalition of bidders* can agree beforehand to collude
    - Propose to artificially lower bids for a good
    - Obtain true value for good
    - Share the profit
  - An auctioneer could employ bogus bidders
    - The bogus bidders could artificially increase bids in open cry auctions
    - Can result in *winners curse*

Combinatorial Auctions

- A combinatorial auction is an *auction for bundles of goods*.
  - Multiple goods are auctioned simultaneously.
  - Each bid may claim any combination of goods.
  - E.g. “I bid $100 for the TV, VCR and couch”.

Combinatorial Auctions

- Define a set of items to be auctioned as:

Set of items for auction

$$\mathcal{Z} = \{Z_1, \ldots, Z_m\}$$

- Given a set of agents $Ag = \{1,\ldots,n\}$, the *preferences* of agent i are given with the *valuation function* opposite:
  - If that sounds to you like it would place a big requirement on agents to specify all those preferences, you would be right.
  - If $v_i(\varnothing) = 0$, then we say that the valuation function for i is *normalised*.
    - i.e. Agent i does not get any value from an empty allocation

Valuation Function

$$v_i : 2^{\mathcal{Z}} \to \mathbb{R}$$

meaning that for every possible bundle of goods $Z \subseteq \mathcal{Z}$, $v_i(Z)$ says how much Z is worth to i.

- Another useful idea is *free disposal*:
  - In other words, an agent is never worse off having more stuff.

Free Disposal

$Z_1 \subseteq Z_2$ implies $v_i(Z_1) \leq v_i(Z_2)$

Allocation of Goods

- An outcome is an allocation of goods to the agents.
  - Note that we don't require all off the goods to be allocated
  - Formally an allocation is a list of sets $Z_1, \ldots Z_n$, one for each agent Ag_i such that $Z_i \subseteq \mathcal{Z}$
  - and for all $i,j \in Ag$ such that $i \neq j$, we have $Z_i \cap Z_j = \emptyset$.
    - Thus no good is allocated to more than one agent.
- The set of all possible allocations of Z to agents Ag is: $alloc(\mathcal{Z}, Ag)$

Maximising Social Welfare

- If we design the auction, we get to say how the allocation is determined.
  - Combinatorial auctions can be viewed as voting methods, with different outcomes relating to different allocations of goods
  - A desirable property would be to maximize social welfare.
    - i.e. maximize the sum of the utilities of all the agents.

- We can define a social welfare function:

$$sw(\underbrace{Z_1, \ldots, Z_n}_{\text{allocations}}, \underbrace{v_1, \ldots, v_n}_{\text{valuations}}) = \sum_{i=1}^{n} v_i(Z_i)$$

Winner Determination

- Given this, we can define a combinatorial auction.
  - Given a set of goods Z and a collection of valuation functions $v_1,\ldots,v_n$, one for each agent $i \in Ag$, the goal is to find an allocation $Z_1^*, \ldots Z_n^*$ that maximises sw:

$$Z_1^*, \ldots, Z_n^* = \underset{(Z_1,\ldots,Z_n)\in alloc(\mathcal{Z},Ag)}{argmax} \quad sw(Z_1, \ldots, Z_n, v_1, \ldots, v_n)$$

- Figuring this out is called the *winner determination* problem.

Winner Determination

- How do we do this?
- Well, we could get every agent i to declare their valuation: $\hat{v}_i$
  - The hat denotes that this is what the agent says, not what it necessarily is.
    - Remember that the agent may lie!
- Then we just look at all the possible allocations and figure out the best one
- One problem here is representation, valuations are exponential: $v_i : 2^{\mathcal{Z}} \to \mathbb{R}$
  - A naive representation is impractical.
    - If there are n goods, we would have to specify 2^n values for each bidder.
- Searching through all possible allocations is **computationally intractable**

Bidding Languages: Atomic bids

- Rather than exhaustive evaluations, allow bidders to construct valuations from the bids they want to mention.
  - *An atomic* bid β is a pair (Z, p) where $Z \subseteq \mathcal{Z}$ is a set of items, and p is a positive real number which indicates the price the relevant bidder is willing to pay for the goods Z
  - A bundle Z' *satisfies* a bid (Z, p) if $Z \subseteq Z'$.

- In other words a bundle *satisfies* a bid if it contains at least the things in the bid.

- Atomic bids *define valuations*

$$v_\beta(Z') = \begin{cases} p & \text{if } Z' \text{ satisfies } (Z, p) \\ 0 & \text{otherwise} \end{cases}$$

- If I make a bid of (Z, p), and you offer me a bundle that satisfies my bid, then my bid says that the amount I am willing to pay is p.

- Atomic bids alone don't allow us to construct many interesting valuations.

Example:

- if my atomic bid is $\boldsymbol{\beta}$ = ({a, b}, 4), i.e., Z = {a, b}, and you offer a bundle Z' = {a, b, c}, then your offer satisfies my bid, since {a, b} ⊆ {a, b, c}.

- a bundle Z' = {b, d} would not satisfy my bid, since Z = {a, b} ⊈ Z' = {b, d};

XOR Bids

- With XOR bids, we specify a number of separate bids but *we pay for at most one of atomic bid*
  - A bid $\beta = (Z_1, p_1)\ XOR\ \ldots\ XOR\ (Z_k, p_k)$ defines a valuation function v_β as follows:

$$v_\beta(Z') = \begin{cases} 0 & \text{if } Z' \text{ does not satisfy any } (Z_i, p_i) \\ \max\{p_i | Z_i \subseteq Z'\} & \text{otherwise} \end{cases}$$

  - I pay nothing if your allocation Z' doesn't satisfy any of my bids
  - Otherwise, I will pay the largest price of any of the satisfied bids.

- XOR bids are *fully expressive*, that is they can express any valuation function over a set of goods.
  - To do that, we may need an exponentially large number of atomic bids.
    - However, the valuation of a given bundle can be computed in polynomial time.

$$\beta_1 = (\{a, b\}, 3)\ XOR\ (\{c, d\}, 5)$$

…"I would pay 3 for a bundle that contains a and b but not c and d. I will pay 5 for a bundle that contains c and d but not a and b, and I will pay 5 for a bundle that contains a, b, c and d..."

From this we can construct the valuation:

$$\begin{aligned} v_{\beta_1}(\{a\}) &= 0 \\ v_{\beta_1}(\{b\}) &= 0 \\ v_{\beta_1}(\{a, b\}) &= 3 \\ v_{\beta_1}(\{c, d\}) &= 5 \\ v_{\beta_1}(\{a, b, c, d\}) &= 5 \end{aligned}$$

XOR Bids

Example: given the XOR bid β_2

$$\beta_2 = (\{e, f, g\}, 2) \text{ XOR } (\{e\}, 2) \text{ XOR } (\{c, d, e\}, 4).$$

This gives us:

$$v_{\beta_2}(\{e\}) = 2$$

$$v_{\beta_2}(\{e, f\}) = 2$$

$$v_{\beta_2}(\{c, d, f, g\}) = 0$$

$$v_{\beta_2}(\{a, b, c, d, e, f, g\}) = 4$$

$$v_{\beta_2}(\{c, d, e\} = 4)$$

OR Bids

- With OR bids, we specify a number of separate atomic bids and *we are prepared to pay for more than one atomic bid.*
  - A bid $\beta = (Z_1, p_1)\ OR\ \ldots\ OR\ (Z_k, p_k)$ defines k atomic bids.
  - You propose an allocation of goods Z', my valuation for Z' is as follows:
    - (1) find the set of atomic bids W contained in β such that every bid in W is satisfied by Z'
    - (2) each pair of bids in W has mutually disjoint sets of goods; i.e. $Z_i \cap Z_j = \varnothing$ for all i,j where $i \neq j$
    - (3) No other subset W' satisfying the above two conditions is better:

$$\sum_{(Z_i,p_i)\in W'} p_i > \sum_{(Z_j,p_j)\in W} p_j$$

$$\beta_1 = (\{a,b\}, 3)\ OR\ (\{c, d\}, 5)$$

"…I would pay 3 for a bundle that contains a and b but not c and d. I will pay 5 for a bundle that contains c and d but not a and b, and I will pay 8 for both bundles that contain a combination of a, b, c and d..."

From this we can construct the valuation:

$$\begin{aligned} v_{\beta_1}(\{a\}) &= 0 \\ v_{\beta_1}(\{b\}) &= 0 \\ v_{\beta_1}(\{a,b\}) &= 3 \\ v_{\beta_1}(\{c,d\}) &= 5 \\ v_{\beta_1}(\{a,b,c,d\}) &= 8 \end{aligned}$$

Note that the **cost of the last bundle is different to that when the XOR bid** was used

OR Bids

(1) find the set of atomic bids W contained in β such that every bid in W is satisfied by Z'

(2) each pair of bids in W has mutually disjoint sets of goods i.e. $Z_i \cap Z_j = \emptyset$ for all i,j where $i \neq j$

(3) No other subset W' satisfying the above two conditions is better:

$$\sum_{(Z_i,p_i)\in W'} p_i > \sum_{(Z_j,p_j)\in W} p_j$$

- Here is another example.
  - $\beta_3 = (\{e, f, g\}, 4)\ OR\ (\{f, g\}, 1)\ OR\ (\{e\}, 3)\ OR\ (\{c, d\}, 4)$

$$\begin{aligned}
v_{\beta_3}(\{e\}) &= 3 \\
v_{\beta_3}(\{e,f\}) &= 3 \\
v_{\beta_3}(\{e,f,g\}) &= 4 \\
v_{\beta_3}(\{b,c,d,f,g\}) &= 4+1=5 \\
v_{\beta_3}(\{a,b,c,d,e,f,g\}) &= 4+4=8 \\
v_{\beta_3}(\{c,d,e\}) &= 4+3=7
\end{aligned}$$

- Z' = {e,f,g} satisfies ({e,f,g},4), ({f,g},1), ({e},3) but W contains only ({e,f,g},4) because of the 2nd and 3rd conditions (or alternatively {e} + {fig})

- Z' = {a,b,c,d,e,f,g} satisfies ({e,f,g},4), ({f,g},1), ({e},3), ({c,d},3) but W contains only {(e,f,g),4} and ({c,d},4) because 2nd and 3rd conditions.

OR Bids

- OR bids are *strictly less expressive* than XOR bids
  - Some valuation functions cannot be expressed
- OR bids also *suffer from computational complexity*
  - Given an OR bid β and a bundle Z', computing $v_\beta(Z')$ is NP-hard

Winner Determination

- Determining the winner in combinatorial auctions is a combinatorial optimisation problem (NP-hard)
  - But this is a worst case result, so it may be possible to develop approaches that are either *optimal* and run well in many cases, or *approximate* (within some bounds).
- Typical approach is to code the problem as an *integer linear program* and use a standard solver.
  - This is NP-hard in principle, but often provides solutions in reasonable time.
  - Several algorithms exist that are efficient in most cases
- Approximate algorithms have been explored
  - Few solutions have been found with reasonable bounds
- Heuristic solutions based on *greedy algorithms* have also been investigated
  - e.g. that try to find the bid with largest price to satisfy, then the next etc

The VCG Mechanism

- Auctions are easy to strategically manipulate
  - In general *we don't know* whether the agents valuations *are true valuations*.
- Life would be easier if they were…
- … so can we make them true valuations?

- Yes!
  - In a generalization of the Vickrey auction.
    - Vickrey/Clarke/Groves Mechanism

- Mechanism is incentive compatible: *telling the truth is a dominant strategy*.

Recall that we could get every agent i to declare their valuation: $\hat{v}_i$

where the hat denotes that this is what the agent says, not what it necessarily is.

- *The agent may lie!*

The VCG Mechanism

- Need some more notation.
  - *Indifferent valuation* function: $\forall Z : v^0(Z) = 0$
    - I.e. the value for a bid that is not interested in the goods
  - Let $Z_1, \ldots, Z_n$ be an allocation of goods Z to agents in *Ag;*
    Let sw_{-i} be the social welfare function without *i*'s benefit:

$$sw_{-i}(Z_1, \ldots, Z_n, v_1, \ldots, v_n) = \sum_{j \in Ag, j \neq i} v_j(Z_j)$$

  - This is how well everyone *except agent i* does from $Z_1, \ldots, Z_n$

- And we can then define the VCG mechanism.

VCG: *How* it works

- Every agent simultaneously declares a valuation v_i
  - Equivalent to one-shot, sealed-bid
  - Remember that it is not necessarily the actual valuation

- The mechanism computes the allocation *maximising social welfare*:

$$Z_1^*, \ldots, Z_n^* = argmax_{(Z_1,\ldots,Z_n)\in alloc(\mathcal{Z},Ag)} sw(Z_1, \ldots, Z_n, \hat{v_1}, \ldots, \hat{v_i}, \ldots, \hat{v_n})$$

- Each agent i pays p_i
  - This is effectively a *compensation* to the other agents for their loss in utility due to i winning an allocation.

VCG: *How* it works

The allocation that maximizes social welfare every agent j declared v_j as its valuation function

$$Z_1^*, \ldots, Z_n^* = \arg \max_{(Z_1,\ldots,Z_n) \in alloc(\mathcal{Z}, Ag)} sw(Z_1, \ldots, Z_n, \hat{v}_1, \ldots, \hat{v}_i, \ldots, \hat{v}_n)$$

The allocation that maximizes social welfare when the agent i declared $v_i = v^0$ as its valuation and other agents declare v_j, $\forall j \neq i$

$$Z_1', \ldots, Z_n' = \arg \max_{(Z_1,\ldots,Z_n) \in alloc(\mathcal{Z}, Ag)} sw(Z_1, \ldots, Z_n, \hat{v}_1, \ldots, v^0, \ldots, \hat{v}_n)$$

The amount agent i has to pay:

$$p_i = sw_{-i}(Z_1', \ldots, Z_n', \hat{v}_1, \ldots, v^0, \ldots, \hat{v}_n) - sw_{-i}(Z_1^*, \ldots, Z_n^*, \hat{v}_1, \ldots, \hat{v}_i, \ldots, \hat{v}_n)$$

- This is the difference in social welfare to agents other than i
  - between the outcome $Z'_1, \ldots, Z'_n$ when i doesn't bid on anything
  - and the outcome $Z^*_1, \ldots, Z^*_n$ when i does participate as normal

VCG: *Why* it works

- With the VCG, each agent pays out the cost (to the other agents) of it having participated in the auction (the reduction in social welfare).
  - *No agent can benefit by declaring anything other than its true valuation*
  - To understand this, think about VCG where there is just a single good to allocate
    - The only agent that pays anything will be the agent i that has the highest bid
    - But if it had not participated, then the agent with the second highest bid would have won
    - Therefore agent i "*compensates*" the other agents by paying this second highest bid

- The result is a dominant strategy for each agent (i.e., using their true valuation when bidding) that guarantees the maximisation of social welfare.
  - i.e. *social welfare maximisation can be implemented in dominant strategies in combinatorial auction.*